KREEDA-ANKANA

Product Requirements Document (PRD) v1.0

Ground & Match Organizer for Village Communities

1. PROBLEM STATEMENT

In rural and semi-urban areas, sports grounds are vital community hubs. However, they suffer from **"Ground Monopolization"** and **"Coordination Gaps."**

- **Current Pain Points:** Informal "first-come, first-served" rules lead to conflicts; smaller or visiting teams cannot find available slots; and match results are lost to word-of-mouth.
- **Target Users:** Village youth, sports club captains, and local ground administrators (Panchayat members).

2. VISION AND GOALS

To digitize the village "Notice Board," transforming unorganized play into a structured sports culture that builds community harmony.

Success Metrics

- **Primary:** 0% double-booking conflicts; Monthly Active Teams (MAT).
- **Secondary:** Challenge Response Rate (number of matches organized via the app); Average app session duration.

3. USERS, ROLES, AND SCENARIOS

Role	Profile	Primary Scenario

Team Captain	Lead of a local Cricket/ Volleyball team.	Checks the calendar at 10 AM, sees a free slot at 5 PM, and books it for his team.
Challenger	Team looking for new opponents.	Posts a public "Open Challenge" for Sunday morning to invite teams from neighboring villages.
Ground Admin	Village authority/ Volunteer.	Updates ground status (e.g., "Ground closed for maintenance due to rain").

4. FEATURES AND REQUIREMENTS

4.1 Ground Calendar & Slot Booking

- **Grid-based View:** Visual representation of 1-hour blocks (6 AM to 8 PM).
- **Conflict Prevention:** Atomic transaction logic; if Team A initiates a booking for 4 PM, that slot is locked for others until the transaction completes or expires.
- **Validation:** Users cannot book more than 2 hours per day to prevent hoarding.

4.2 Challenge Board

- **Real-time Updates:** Instant push via Firebase when a new challenge is posted.
- **Reply Threading:** Interested teams can "Comment" or "Accept" the challenge directly within the post.

4.3 Score Wall

- **Social Proof:** A feed showing "Team A beat Team B (21-18, 21-15)".
- **Persistence:** Scores are saved locally via Room and synced to the cloud when online.

5. USER FLOWS AND WIREFRAMES

Booking Flow

Home -> Select Date -> Tap Empty Slot -> Enter Team Name -> Confirm -> Slot turns Red (Occupied).

> ***Wireframe Note:*** *Use a high-contrast Grid. Occupied slots should show the Team Logo/Name. The 'Book' button should be a floating action button (FAB) with a sporty icon.*

6. TECHNICAL ARCHITECTURE

- **Frontend:** Kotlin with **Jetpack Compose** (Declarative UI). MVI (Model-View-Intent) for state management.
- **Backend:**
  - **Firebase Realtime DB:** Low-latency data for the Challenge Board.
  - **Firestore:** Structured data for Team Profiles and Historical Scores.
- **Local Persistence: Room DB** acting as the Single Source of Truth for the UI.
- **Security:** Firebase Phone Auth (OTP) to ensure accountability.

7. DATA MODEL (SAMPLE SCHEMA)

```
Entity: Booking
{
  bookingId: String (PK),
  groundId: String (FK),
  teamId: String,
  startTime: Long (Timestamp),
  endTime: Long (Timestamp),
  status: "CONFIRMED" | "CANCELLED"
}
```

8. MILESTONES AND PHASES

Phase 1: MVP (Weeks 1-4)

- Core Calendar and Slot Booking logic.
- Basic Firebase integration.
- Static Score Wall.

Phase 2: Social & Growth (Weeks 5-8)

- Real-time Challenge Board replies.
- Push notifications for match reminders.
- Automated "Team Leaderboards."

9. RISK ASSESSMENT

- **Technical:** Connectivity issues in deep rural areas. *Mitigation:* Robust Room DB offline-first implementation.
- **Adoption:** Dominance by one powerful group. *Mitigation:* Admin "Super-User" role to moderate bookings.